UART Communication NPET

last rev. 03.01.2019

Firmware version: Firmware 03012019_ADI_v2

Initial serial interface settings:

Baud rate: 115200
Data bits: 8
Stop bit: 1
Parity: none
Handshake: No

Commands structure:

Commands consists of 1 ASCII character and unsigned int number. They are terminated by LF or CR LF. Commands which do not require fast measurement response are echoed back. All messages received from NPET are terminated by CR LF excepts binary measured data transfer. If the command was not excepted the NPET will return "Bad Command!\r\n". The time of arrivals (epochs) are transfer in ASCII (after power on) or binary format.

Measured data in ASCII format:

The epochs consists of three numbers separated by space and terminated by CR LF, e.g. "X Y Z\r\n"
Time of arrival (ToA) is computed according to

$$ToA[s] = \frac{X*2^{30}+Y}{100\,e^6} + \frac{Z*2^{-16}}{100\,e^6}.$$

Measured data in binary format:

The binary format consists of 13 unsigned char (8bits) number. Each measurement starts with hexadecimal 0x01 and ends with XOR check sum. The detailed message is in table below.

0x01	No. of bytes	Packet No.	uc(1)	uc(2)	uc(3)	uc(4)	uc(5)	uc(6)	uc(7)	uc(8)	uc(9)	XOR

No. of bytes – is the length of one measurement message (Packet No. + uc(1) + … + uc(9) + XOR = 11).
Packet No. - is free overflowing 8bit unsigned char which counts message numbers.
uc(1) – uc(9) – measured data
XOR - check sum; is 8bit unsigned char computed from a whole message
Time of arrival (ToA) is computed according to

$$ToA[s] = \frac{(uc(1)+uc(2)*2^8+uc(3)*2^{16})*2^{24}}{100e6} + \frac{(uc(4)+uc(5)*2^8+uc(6)*2^{16})}{100e6} + \frac{(uc(7)+uc(8)*2^8+uc(9)*2^{16})-(2^{22}-1)}{2^{16}*100e6}$$

The maximum repetition rate of sending of ToAs is 830Hz for baud rate 115200.

Command list:

Note:

If the pulse generator is set to generate pulses (e.g. p9999) and command "i" or "d" is set. Then no pulse will be generated and NPET will be in state of waiting for incoming pulse. To avoid this situation first switch off generating pulses (e.g. p0000) and than use command "i" or "d".

<table>
<tr><th>Command</th><th>uint lim.</th><th>Command description</th><th>echoed</th></tr>
<tr><td>p<uint></td><td>0 – 999999999</td><td>Set NPET to start generating uint pulses of preset frequency; if uint = 9999 then never ending pulse generation starts</td><td>yes</td></tr>
<tr><td>k<uint></td><td>0 – 1000000</td><td>Set NPET frequency to uint in Hz of generating pulses. Initial value = 100 Hz</td><td>yes</td></tr>
<tr><td>e<uint>, h<uint></td><td>1 - 999999999</td><td>Set NPET into measurement mode and measure uint ToA; e – channel IN1 is activated; h – channel IN2 is activated. If uint = 9999 set NPET in to streaming mode.
When no pulses are received NPET will be in wait state and will response only to command s. To break this wait state send “c\n” or “c\r\n”.</td><td>no</td></tr>
<tr><td>i<uint>, d<uint></td><td>1 – 999999999</td><td>Set NPET to start generating uint pulses of preset frequency and perform uint number of measurements of ToA. The main purpose of this command is TWTT. i – channel IN 1 is activated; d – channel IN 2 is activated. If uint = 9999 set NPET in to streaming mode.
When no pulses are received NPET will be in wait state and will response only to command s. To break this wait state send “c\n” or “c\r\n”.</td><td>no</td></tr>
<tr><td>s<uint></td><td>any value less then 999999999</td><td>Return status of NPET. One byte unsigned character.<table><tr><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr><tr><td>NA.</td><td>NA.</td><td>NA.</td><td>NA.</td><td>NA.</td><td>'0' Binary data; '1' ASCII</td><td>'0' IN1, '1' IN2</td><td>'1' waiting for coming pulses</td></tr></table></td><td>no</td></tr>
<tr><td>r<uint></td><td>1</td><td>Resets raw counter.</td><td>yes</td></tr>
<tr><td>a<uint></td><td>0,1</td><td>uint = 0; measured data are in binary format
uint = 1; measured data are in ASCII format (default after powering NPET on)</td><td>yes</td></tr>
<tr><td>w<uint></td><td>4800, 9600, 19200, 38400, 57600, 115200, 230400, 460800, 576000, 921600, 2000000, 5000000</td><td>Change baud rete according to uint; allowed values are (4800, 9600, 19200, 38400, 57600, 115200, 230400, 576000, 2000000, 5000000).
Baud rates 2000000 and 5000000 are supported only on pins TST7_N (Rx) and TST6_N (Tx). In this configuration pin TST2_N must be shorted to gnd.
Initial value after powering NPET on is 115200</td><td>yes</td></tr>
<tr><td>j<char[max 30]></td><td></td><td>Saves user constant of max length 30 characters terminated by LF or 29 characters terminated by CR LF.</td><td>yes</td></tr>
<tr><td>n<uint></td><td>1</td><td>Get user saved constant. If no constant was saved then return “\r\n”.</td><td>yes</td></tr>
<tr><td>g</td><td>–</td><td>Read temperatur, answer Temp = xx.xxx</td><td></td></tr>
<tr><td>c</td><td>--</td><td>-Returns “c0\r\n” when NPET is reddy to receive commands
-Breaks the streaming modes of reading ToA, returns “c1\r\n”</td><td>(yes)</td></tr>
<tr><td>?</td><td>–</td><td>Prints firmware version</td><td>--</td></tr>
</table>

Baud rate is generated according to:
80MHz/(n*16)
where
n = 1, 2, 4,

Baud rate	Error rate
115200	1.3%
230400	1.3%
460800	1.4%
555555	error rate 3.5% for 576000
576000	Represent exactly 625000, at 576000 is error rate 8.5%
921600	8.5%

Debug only command list:

Command	**uint** lim.	Command description	echoed
m<uint>		tart measurement and send raw data without raw counter; **uint** is number of measurement. Data are sent in ASCII format; raw reading from ADC; do not depends on binary transfer off/on; **uint** = 9999 never-ending loop	no
t<uint>		start measurement without excitation send raw data; **uint** is number of measurement. Data are sent in ASCII format; raw reading from ADC; do not depends on binary transfer off/on; **uint** = 9999 never-ending loop	
o<uint>		start measurement and send raw data of X1 and X2 without raw counter; **uint** is number of measurement. Data are sent in ASCII format; raw reading from ADC; **uint** = 9999 never - ending loop. Data are sent in ASCII format if a1 command was set or binary format if a0 was set; binary format: Sends 32+32 samples each sample is 16 bits long; sending order YX where Y – MSB; X – LSB	
l<uint>		sets AGC coefficient must be in form floor((float)AGC*32)	
f<uint>		- **uint = 0 –** set ON filtr X2 - **uint = 1 –** set OFF filtr X2 - **uint = 2** – set AGC ON - **uint = 3** – set AGC OFF - **uint = 4** – set ON filtr X2 and AGC - **uint = 5** – set OFF filtr X2 and AGC	
v<uint>		start measurement and send vernier only in ASCII format	no
u<uint>		set p9999 and start infinity measurements which are averaged by uint	